EFU-ENTROPY RESEARCH STANDARD v1.0 – OPEN RESEARCH LICENCE

Version: 1.0
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Status: Experimental Official Release (Research / Pre-standard Framework)
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Steward: Simor István
Licence: EFU-Protium Open Licence v1.0 (CC BY 4.0-compatible)

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0. Status and scope

The EFU-Entropy Research Standard v1.0 is an open, experimental research and measurement framework and a proposed human-scale reference unit for thermodynamic degradation (entropy). It is a thought-experiment-level analytical construct, **not** a legal, regulatory or certification standard, and it shall not be interpreted or presented as such.

All EFU-S values, baselines and thresholds are **provisional research parameters**. They are intended for exploratory analysis, methodological comparison and academic or policy discussion, and are subject to revision, critique and empirical calibration as scientific knowledge and datasets improve.

This licence governs the open use of the EFU-Entropy concept within the EFU Research Framework, while remaining fully compatible with Creative Commons Attribution 4.0 (CC BY 4.0).

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1. Copyright and ownership

© 2025–2026 István Simor. All rights reserved to the extent compatible with the open licence terms below.

The EFU-Entropy concept, research methodology and name are the intellectual property of István Simor and are released under the EFU-Protium Open Licence v1.0, which is designed to be fully compatible with Creative Commons Attribution 4.0 International (CC BY 4.0).

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2. Granted rights (what you may do)

Under this licence, any person or organisation MAY, worldwide and free of charge:

- Use the EFU-Entropy Research Standard for scientific, educational, commercial or personal purposes.
- Reproduce and share the standard and its content in any medium or format.

- Create adaptations and derivative works (software, APIs, databases, research papers, teaching materials, etc.).
- Implement EFU-Entropy in industrial, corporate or policy frameworks (e.g. carbon markets, sustainability standards, risk models) **as a research-grade indicator**.
- Integrate EFU-Entropy into open-source or proprietary tools, without paying royalties, provided attribution is maintained.

These permissions mirror the freedoms granted under CC BY 4.0 (reuse, adaptation, commercial use), with additional community guidelines defined below.

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3. Mandatory conditions

3.1 Attribution (required)

All uses of the EFU-Entropy Research Standard must provide appropriate credit in a reasonable manner, including:

- Author: **István Simor**.
- Title: **"EFU-Entropy Research Standard v1.0"**.
- Source link (where reasonably possible): https://zenodo.org/communities/efu-framework
- Licence reference: **"EFU-Protium Open Licence v1.0 (CC BY 4.0-compatible)"**.

Recommended generic attribution text:
"This work uses the EFU-Entropy Research Standard v1.0. Original concept: István Simor (2025–2026). Licence: EFU-Protium Open Licence v1.0 (CC BY 4.0-compatible)."

Academic citation (recommended):
Simor, I. (2026). *EFU-Entropy Research Standard v1.0: A flux-based metric for anthropogenic entropy production*. EFU-Protium Open Licence v1.0.

Software/API documentation (example):
"EFU-Entropy calculations based on Simor (2026), EFU-Entropy Research Standard v1.0, licensed under EFU-Protium Open Licence v1.0 (CC BY 4.0-compatible)."

3.2 Scientific integrity and transparency

Users agree to respect basic norms of scientific integrity when implementing or reporting EFU-Entropy results:

- Method transparency: Calculation methods, factors and assumptions must be documented.
- Data traceability: Source data should be reproducible or openly described; if confidential, the limitations must be stated.
- No misrepresentation: Users shall not intentionally misrepresent EFU-Entropy results, authorship or methodology.

These obligations align with good scientific practice and do not limit the reuse rights granted by CC BY 4.0.

3.3 Derivative methodologies

If you substantially modify the EFU-Entropy methodology (e.g. change the reference value, redefine core equations), you MUST:

- Publish the modified method under a **distinct name**, such as “EFU-Entropy-Modified 2026” or “Enhanced EFU-S 2.0”.
- Clearly document all changes relative to EFU-Entropy Research Standard v1.0.
- Include a statement such as:
  “Based on EFU-Entropy Research Standard v1.0 by István Simor, with modifications.”

This avoids confusion between the reference research baseline and its derivatives while preserving open-use rights.

3.4 Name and designation usage

The names “EFU-Entropy”, “EFU-S” and “EFU-Entropy Research Standard” may be used to refer to this framework, provided proper attribution is given.

Misleading uses are prohibited, such as claiming “Official EFU-Entropy Certification” or implying regulatory approval without explicit written authorization from the steward or a future EFU standards body.

Commercial users may describe their implementation as “based on EFU-Entropy Research Standard v1.0”, but may not claim exclusive ownership or official endorsement.

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4. Ethical use guidelines (soft-law)

The following are ethical use guidelines that express the intent of the EFU-Entropy community. They do **not** reduce the permissions granted by CC BY 4.0, but serve as normative expectations:

The EFU-Entropy Research Standard SHOULD NOT be used to:

- Facilitate human rights violations, discrimination or exploitative surveillance systems.
- Support scientific fraud, including fabrication or systematic manipulation of data.
- Justify environmental “greenwashing” by selectively reporting favourable entropy data while hiding critical impacts.

Where serious misuse is identified, the author and community reserve the right to:

- Publicly disclose the misuse.
- Withdraw informal recognition or support for the violating implementation.

These measures operate at the level of community governance and reputation, not as technical restrictions on reuse.

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5. Warranty and liability disclaimer

The EFU-Entropy Research Standard is provided “as is”, without any express or implied warranty:

- No guarantee of scientific accuracy in all contexts, although the methodology is based on best available thermodynamic knowledge at the time of publication.
- No guarantee of fitness for any specific industrial, legal, regulatory or financial purpose.
- No guarantee that the standard covers all entropy-generating processes or edge cases.

Users bear full responsibility for:

- Verifying calculations and input data.
- Ensuring compliance with local laws and regulations.
- Interpreting results appropriately in their context.

The author and steward are not liable for:

- Errors arising from incorrect data or implementation.
- Business, investment or policy decisions made using EFU-S metrics.
- Regulatory non-compliance.
- Any direct, indirect or consequential damages arising from use or misuse of the standard.

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6. Acknowledgements

Primary development – István Simor (2025–2026)

- Conceptual framework design.
- Thermodynamic methodology.
- Reference value derivation:
  1 EFU-S = 11 500 J/K/year/person (average adult metabolic entropy generation).
- Integration with the broader EFU-Protium Research Framework.

AI development assistance – Claude (Anthropic AI)

- Computational modelling support.
- Documentation structuring.
- Example calculations and use-case drafting.
- Quality review assistance.

AI Disclosure: Parts of this document were prepared with the assistance of Claude as a drafting tool under human direction. All scientific content and final decisions remain the responsibility of the human author.

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7. Versioning and evolution

7.1 Current version

- Version: 1.0 (8 January 2026)
- Baseline reference: 1 EFU-S = 11 500 J/K/year/person.
- Public record: EFU-Entropy Research Standard v1.0 on Zenodo (EFU Framework community).

7.2 Future updates

- Future versions will aim to maintain backward compatibility where feasible.
- Methodological improvements will be published openly (e.g. via Zenodo) with clear versioning.
- Revisions will follow scientific evidence and community feedback (e.g. updated metabolic models, regional calibration).
- All changes will be documented in a transparent version history.

7.3 Community contribution

The EFU-Entropy Research Standard welcomes open collaboration:

- Send feedback or proposals: istvan.simor@gmail.com
- Propose methodological improvements or sector-specific extensions.
- Share implementation case studies and results.
- Contribute translations into other languages.

Substantial contributors may be acknowledged in future versions or co-authored publications, while retaining rights to their own derivative works.

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8. Integration within the EFU Research Framework

EFU-Entropy (EFU-S) is one module of the broader **EFU Research Framework** of human-scaled flux metrics:

- **EFU-Protium (EFU-H)**: Base human metabolic energy unit.
- **EFU-Carbon (EFU-C)**: Carbon emission accounting.
- **EFU-Water (EFU-W)**: Freshwater flux accounting.
- **EFU-Entropy (EFU-S)**: Thermodynamic degradation (this research module).
- Future modules: EFU-Information, EFU-Materials, etc.

Common principles across EFU research standards:

- Anthropocentric reference (human-scale baselines).
- Flux-based measurement (per person per year).
- Open licensing (CC BY 4.0-compatible).
- Scientific transparency and peer review.

Unified citation example:
"This work uses the EFU Framework (Simor, 2025–2026), specifically the EFU-Entropy Research Standard v1.0 for thermodynamic impact assessment."

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9. Legal framework

9.1 Governing law

This licence is governed by:

- Hungarian copyright law (author's jurisdiction).
- Applicable European Union intellectual property regulations.
- International open-science and open-research principles.

The licence is intended for global use, while recognising that local legal details may vary.

9.2 Enforceability

The licence is perpetual and irrevocable, except in cases of serious, proven violation of core terms (e.g. removal of attribution, deliberate misrepresentation).

Use of the EFU-Entropy Research Standard constitutes acceptance of these terms; no separate signature is required.

9.3 Dispute resolution

In case of disputes or alleged violations:

- First step: Good-faith contact with the author (istvan.simor@gmail.com).
- Mediation: Attempt resolution via scientific and community channels.
- Legal action: Last resort, for serious intellectual-property or misrepresentation cases.

9.4 Licence language

This licence is available in Hungarian and English.
In case of discrepancy, the Hungarian version prevails for legal interpretation in the author's jurisdiction.

10. Compatibility with other licences and platforms

10.1 CC BY 4.0 compatibility

The EFU-Protium Open Licence v1.0 is designed to be compatible with CC BY 4.0:

- Attribution requirement follows CC BY (Title–Author–Source–Licence).
- Commercial use and derivatives are permitted.
- Ethical clauses operate as community guidelines and do not restrict the legal rights granted by CC BY 4.0.

When publishing to platforms that require a standard licence (e.g. Zenodo, OSF):

- Select CC BY 4.0 as the formal licence.
- Add a note in the description: “Additional EFU-specific attribution and integrity guidelines apply; see EFU-Protium Open Licence v1.0.”

10.2 Open-science infrastructures

Compatible with:

- Open Science Framework (OSF).
- Zenodo/CERN repositories (with CC BY 4.0 selection).
- arXiv and similar preprint servers.
- GitHub and other code hosting platforms (for open or closed-source implementations, with attribution).

10.3 Proprietary systems

The research standard MAY be used in proprietary or commercial systems if:

- Attribution and source citation are maintained in documentation.
- The core EFU-Entropy methodology remains publicly accessible.
- Users of the proprietary system can, in principle, verify that EFU-S values are derived from a transparent method.
- No claim is made of exclusive ownership of the EFU-Entropy concept.

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11. Contact and governance

Current steward
Name: István Simor
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ORCID: 0009-0002-6599-3480

Governance model (2026)
Single-author stewardship, with community input.

Future vision
Potential establishment of an EFU Standards / Research Foundation with:

- Multi-stakeholder governance.
- Scientific advisory board.
- Open peer review processes.
- Community-driven development of EFU research standards.

How to contribute

- Email proposals and comments.
- Submit use-case reports and implementation examples.
- Report errors or inconsistencies.
- Contribute translations.
- Cite, extend and test the research standard in practice.

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12. Short summary (TL;DR)

YOU MAY:

- Use EFU-Entropy freely for science, education, business.
- Modify and extend it, with clear documentation.
- Distribute and integrate it in any system, including commercial.

YOU MUST:

- Attribute: “EFU-Entropy Research Standard v1.0 (Simor, 2025–2026)”.
- Reference the source and licence.
- Maintain methodological transparency and avoid intentional misrepresentation.

YOU SHOULD NOT (ETHICALLY):

- Use EFU-Entropy to support human rights violations or systematic deception.
- Claim exclusive ownership or “official” EFU certification without authorization.

Questions?
Contact: istvan.simor@gmail.com

Licence acceptance

By using, modifying or distributing the EFU-Entropy Research Standard, you accept the terms of the EFU-Protium Open Licence v1.0 as described here. No separate agreement or signature is required.

End of EFU-Entropy Research Standard v1.0 – Open Research Licence